Opinion on
"The Quality of Lattice Sequences for Arbitrary Sample Size"

Summary of the paper

The paper studies extensible rank-1 lattice sequences in base b and the equal-weight cubature rule based on the first n points, where n is not necessarily a power of b. The main results are:

- Theorem 1(a): for arbitrary n, the worst-case error $P_\alpha(\boldsymbol{h}, \boldsymbol{\gamma}, n, s)$ is of order n^{-1} (up to logarithmic factors).
- Theorem 1(b): for $n = \lambda b^p$ with fixed λ, the worst-case error is of order $n^{-\alpha}$ (up to logarithmic factors and a factor $\lambda^{\alpha-1}$).
- Section 5: in a Hilbert space setting, optimal (non-equal) sample weights are computed and shown numerically to reduce the error.

Theorem 2 gives analogous dimension-independent bounds under summability conditions on the coordinate weights $\boldsymbol{\gamma}$.

Overall assessment

The main argument (splitting the first n points into shifted copies of the lattices with b^ℓ points and applying the triangle inequality) is standard and appears to be correct. However, the contribution is thin:

- the new bounds are *weaker* than what follows immediately from the known result for $n = b^m$ by simply ignoring the surplus points;
- the part on optimal weights illustrates a fact that holds trivially.

Details are given below. Equation numbers such as (9), (12), (14) refer to the paper. Throughout, we write

$$L(n) := \max(1, (\log n)^{\alpha(s+1)}) \left[\max(1, \log\log(n+1))\right]^{\alpha(1+\epsilon)}$$

for the logarithmic factor used in the paper, so that the known bound (9) from [HN03] reads

$$P_\alpha(\boldsymbol{h}, \boldsymbol{\gamma}, b^m, s) \le C_P\, b^{-m\alpha}\, L(b^m), \qquad m = 0, 1, 2, \ldots$$

1 Theorem 1 compared with the known result for $n = b^m$

Let $n \in \mathbb{N}$ be arbitrary and $m := \lfloor \log_b n \rfloor$, so that $b^m \le n < b^{m+1}$. Simply discarding the surplus points, i.e., using only the first b^m points with equal weights, gives by (9) the error bound

$$C_P\, b^{-m\alpha}\, L(b^m) \;\le\; C_P\, b^\alpha\, n^{-\alpha}\, L(n) \qquad \text{for every } n \in \mathbb{N}. \tag{T}$$

So the optimal order $n^{-\alpha}$ (up to logarithmic factors) is already available for *all* n, with the same generating vector $\boldsymbol{h}$.

Note that (12) holds simultaneously for all λ and p with the same $\boldsymbol{h}$, i.e., it bounds the error of one fixed point sequence for every n, and it gives the order $n^{-\alpha}$ along any subsequence with bounded λ. However, (T) does the same, and it is always better: for $n = \lambda b^p$, let $k := \lfloor \log_b \lambda \rfloor$, so that $m = p + k$ and the left-hand side of (T) equals $C_P\, (\lambda / b^k)^\alpha\, n^{-\alpha} L(b^m)$. Since

$$\lambda^{\alpha-1}\, \frac{b-1}{1-b^{1-\alpha}} \;>\; \Big(\frac{\lambda}{b^k}\Big)^\alpha \qquad \text{for all } b \ge 2,\ \alpha > 1,\ \lambda \in \mathbb{N}$$

and $L(b^m) \le L(n)$, the bound (T) is strictly better than (12) for all b, α and λ. In other words, the new bound (12) for the rule with all n points is always weaker than the known bound for the first $b^m \le n$ points. (Both are upper bounds, but the paper gives no indication that using all n points actually helps.)

Similarly, the $O(n^{-1})$ bound of Theorem 1(a) for arbitrary n is essentially [HKKN12, Corollary 1], with the logarithmic factor traded for a larger constant. Compared with (T), it only shows that equal weights on all n points are worse than discarding the surplus points.

2 Optimal sample weights (Section 5 of the paper)

Let $m := \lfloor \log_b n \rfloor$. The weight vector

$$\boldsymbol{w} = (\underbrace{b^{-m}, \ldots, b^{-m}}_{b^m \text{ entries}}, \underbrace{0, \ldots, 0}_{n-b^m \text{ entries}})$$

satisfies $\boldsymbol{w}^\top \mathbf{1} = 1$ and is therefore admissible in the minimization problem of Section 5.1 of the paper. Consequently,

$$P_{\alpha,2}(\boldsymbol{h}, \boldsymbol{\gamma}, n, s, \boldsymbol{w}_{\text{opt}}) \;\le\; P_{\alpha,2}(\boldsymbol{h}, \boldsymbol{\gamma}, b^m, s, b^{-m}\mathbf{1}).$$

By the same argument applied to any $n' \le n$ (the point sets are nested), the map $n \mapsto P_{\alpha,2}(\boldsymbol{h}, \boldsymbol{\gamma}, n, s, \boldsymbol{w}_{\text{opt}})$ is non-increasing. Thus the observations that the curve for optimal weights in Figure 6 is non-increasing and that the ratios in Figure 7 stay at or below one are automatic one-line consequences, not numerical findings. In particular, optimal weights trivially achieve the same order of convergence for all n as for $n = b^m$ (up to a constant factor).

Two further points:

- The experiments use $\alpha = 1$. For $\alpha = 1$, equal weights already achieve order n^{-1} (up to logarithmic factors) for all n [HKKN12, Corollary 1], so there is no rate to be recovered. The interesting regime $\alpha > 1$, where equal weights do lose the rate, is not tested. Moreover, $\alpha = 1$ lies outside the range $\alpha > 1$ of Theorems 1–3.

- The question of non-equal weights that yield higher order convergence for all N is the subject of [HKKN12]. The relation between Section 5 and that paper should be discussed.

3 Further remark

Section 5.1 states that the Gram matrix K is Toeplitz. This seems to be wrong in general: in van der Corput order, $\boldsymbol{x}_i - \boldsymbol{x}_j$ is not a function of $i - j$ (for $b = 2$: $\phi(1) - \phi(0) = 1/2$, but $\phi(2) - \phi(1) = -1/4$). Hence the claimed cost $O(sn + n^2)$ does not follow. (For $n = b^m$ there is a circulant structure after reordering the points, but not for general n.)

4 What would make the paper worthwhile

The paper would become a useful contribution if it went beyond the trivial baseline (T), i.e., if it showed that the information in the surplus points can actually be exploited. Possible routes are:

1. a bound for the equal-weight rule with all n points (e.g. for $n = \lambda b^p$) that provably improves on (T); this probably requires exploiting cancellations between the shifted blocks rather than the triangle inequality. Alternatively, a lower bound that clarifies when using all points is better or worse than disregarding surplus points.

2. For weighted rules, explicit and cheaply computable weights using all n points, together with a theorem showing a better bound than (T) for such rules, thus removing the factor $(n/b^m)^\alpha \le b^\alpha$ lost by truncation; and a clear comparison with the weighted compound rules of [HKKN12].

3. Numerical experiments for $\alpha > 1$ (e.g. $\alpha = 2$) that include the truncated rule and the rules of [HKKN12] as baselines, for instance by plotting the error of each method relative to that of the truncated rule. In any case, the baseline (T) and the trivial monotonicity of optimal weights should be stated explicitly, and the cost analysis in Section 5.1 should be corrected.

A revised version along these lines could be of genuine interest, in particular for practitioners who face arbitrary sample sizes.

References

[HN03] F. J. Hickernell, H. Niederreiter, The existence of good extensible rank-1 lattices, *J. Complexity* 19 (2003), 286–300.

[HKKN12] F. J. Hickernell, P. Kritzer, F. Y. Kuo, D. Nuyens, Weighted compound integration rules with higher order convergence for all N, *Numer. Algorithms* 59 (2012), 161–183.